

Study Planning Agent

Local Intelligent Software Agent Prototype for COMPSCI 767 Assignment 2

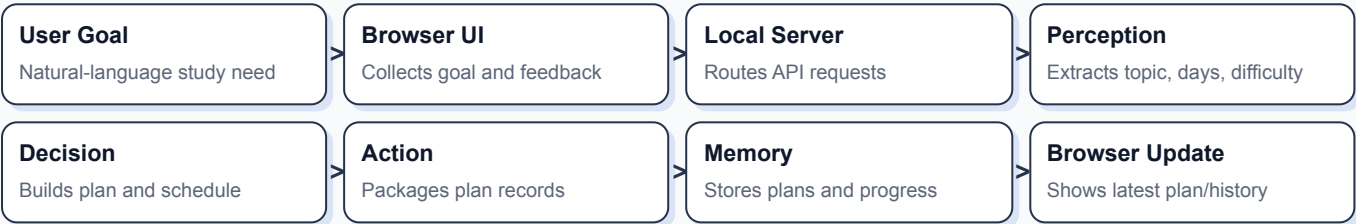
GitHub repository: <https://github.com/Leth3323/A-local-intelligent-software-agent-prototype-for-COMPSCI-767-Assignment-2>

Project Overview

Study Planning Agent is a local browser-based intelligent software agent prototype. It acts as a personal study planning assistant: the user enters a natural-language goal, and the agent converts it into a structured plan with a concise title, topic, duration, difficulty level, task type, summary, strategy, priorities, notes, and daily tasks.

The prototype stores plans locally and supports practical interaction after generation, including viewing plan details, updating a plan from feedback, deleting previous plans, and changing task progress between **To Do**, **In Progress**, and **Done**.

System Design and Agent Workflow



Agent Modules

Perception Interprets the raw user goal and identifies the topic, intended duration, difficulty level, and task type.	Decision Creates or revises the structured study plan, including daily tasks, strategy, priorities, warnings, and notes.
Action Packages the generated output into a plan record and preserves task-status information during plan updates.	Memory Saves generated plans, feedback revisions, task progress, and deletions in the local project data store.

How the Agent Works

The user submits a goal through the browser. The local server passes the request to the agent pipeline, which analyses the input, generates a day-by-day plan, records it locally, and returns it to the interface. The user can then inspect details, mark progress, revise the plan with feedback, or remove outdated records.

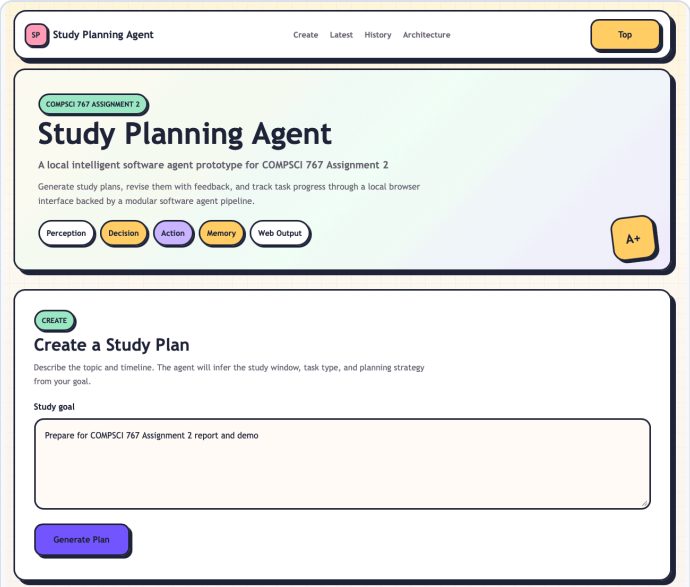


Figure 1. Home page and goal input interface. The user enters a natural-language study goal, and the agent receives this input as the starting point for plan generation.

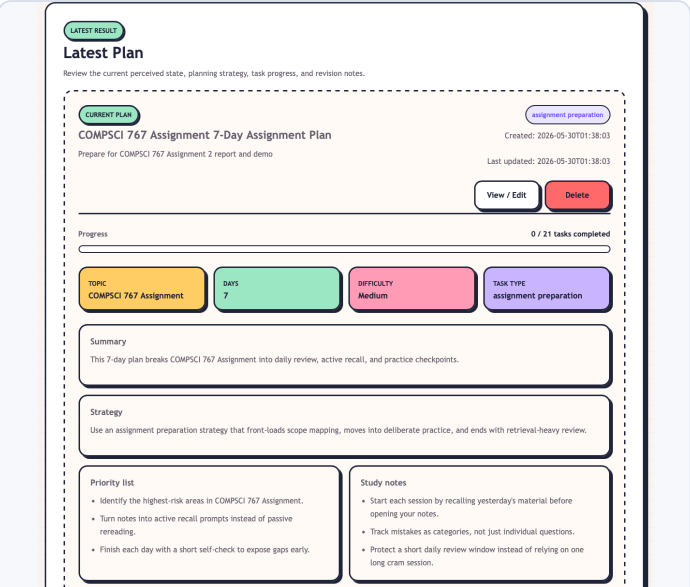


Figure 2. Generated plan overview. The agent analyses the user goal and converts it into a structured study plan, including a concise title, topic, duration, difficulty level, task type, summary, strategy, priorities, and study notes.

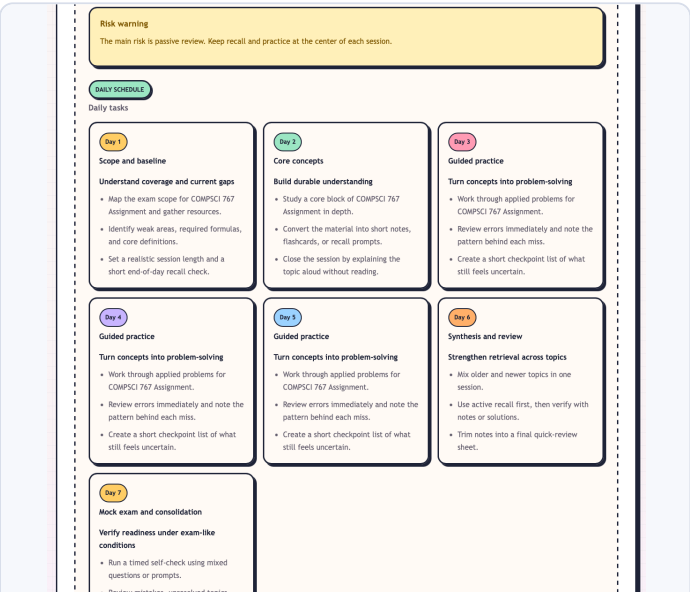


Figure 3. Daily task schedule. The agent breaks the goal into actionable daily tasks, showing how the planning output is organised into smaller steps for the user to follow.

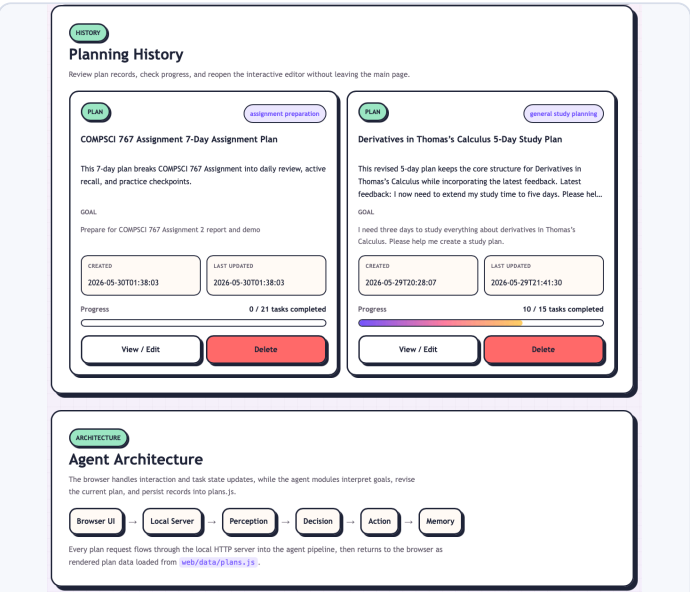


Figure 4. Planning history and agent architecture. The system stores previous plans, tracks task progress, supports view/edit and delete actions, and presents the agent workflow from browser input to perception, decision, action, and memory.

Conclusion

The prototype demonstrates a complete local intelligent-agent loop. It turns informal study goals into actionable plans, stores the results locally, and supports continued user interaction through feedback, task-progress updates, history, editing, and deletion.